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Impact of Pitching Angle and Frequency on Flapping Foil Hydrodynamics: A CFD Study with Experimental Validation

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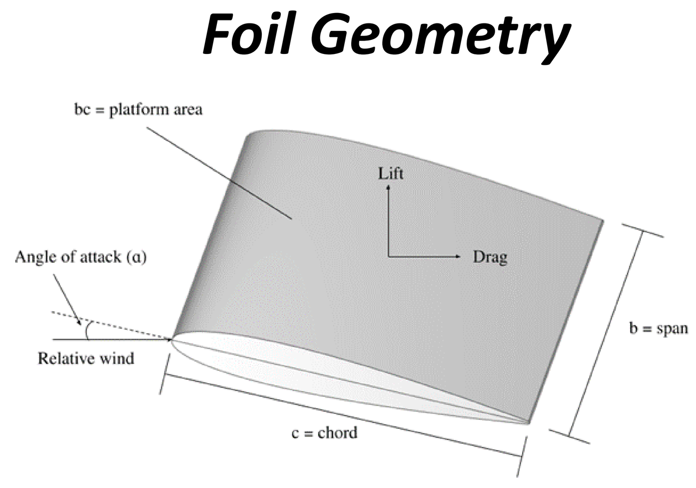
Abstract

Flapping foils inspired by the movement of aquatic creatures offer excellent efficiency for underwater propulsion. The key parameters such as pitching amplitude, frequency, and phase difference, need to be analyzed to optimize its hydrodynamic performance. This work investigates the effects of varying pitching angles and frequencies on the lift and thrust generated by flapping (symmetric and asymmetric) foils using Computational Fluid Dynamics (CFD) and validates the results with experimental testing. The results showed a strong correlation between specific pitching angles and frequencies with improved thrust and lift, supporting their application in bioinspired underwater robotics and energy harvesting.

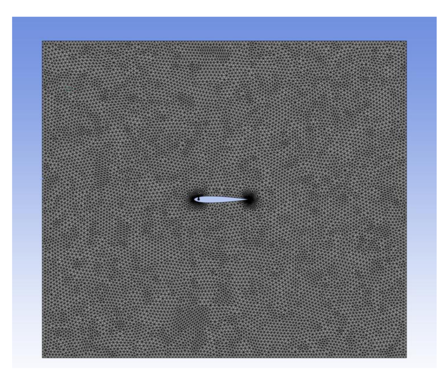
Methodology and Experimental Setup

CFD Simulation steps

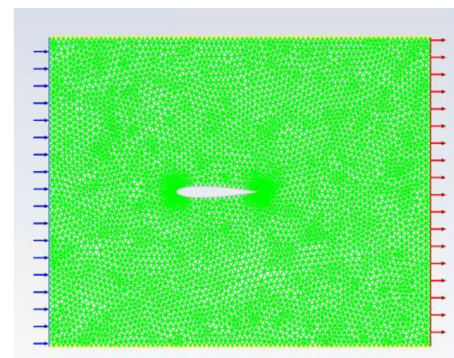
Foil Geometry



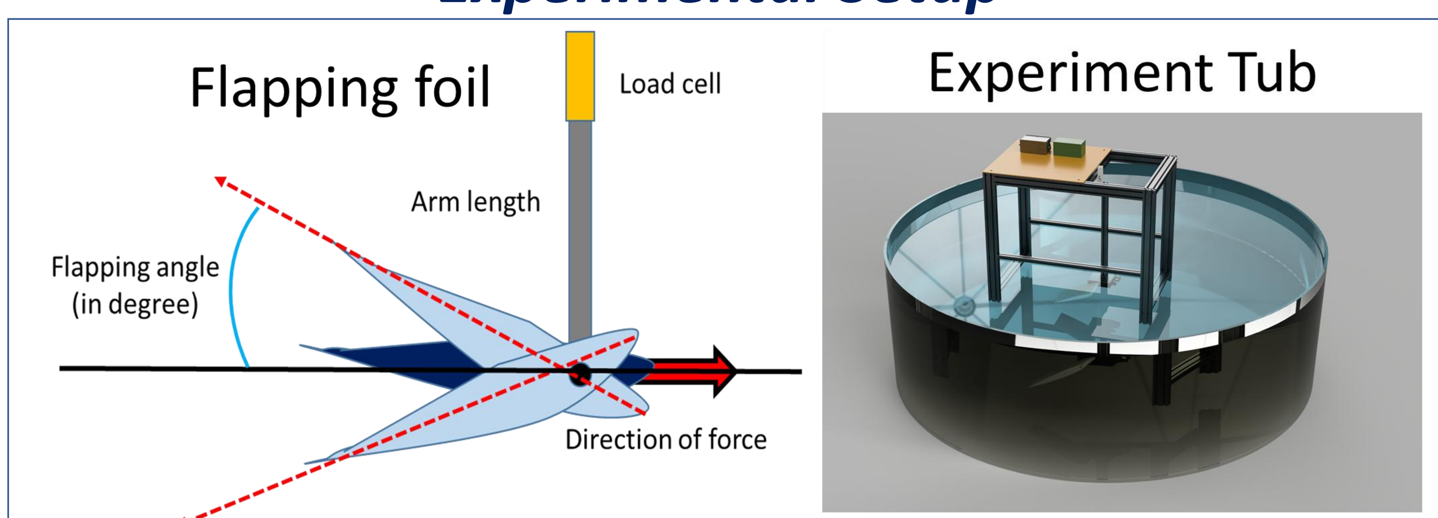
Fine Mesh Around Geometry



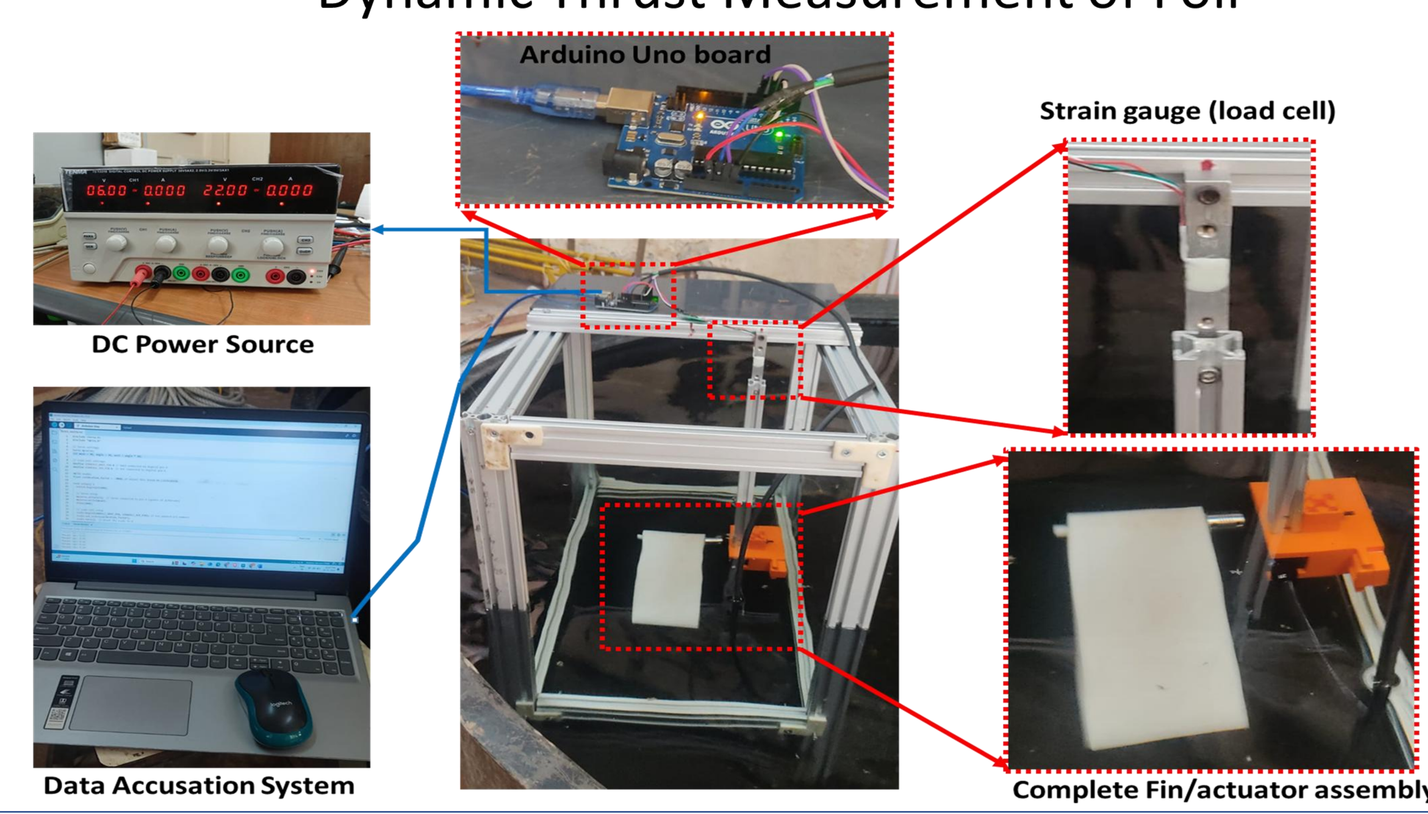
Fluent Setup with Boundary Condition



Experimental Setup

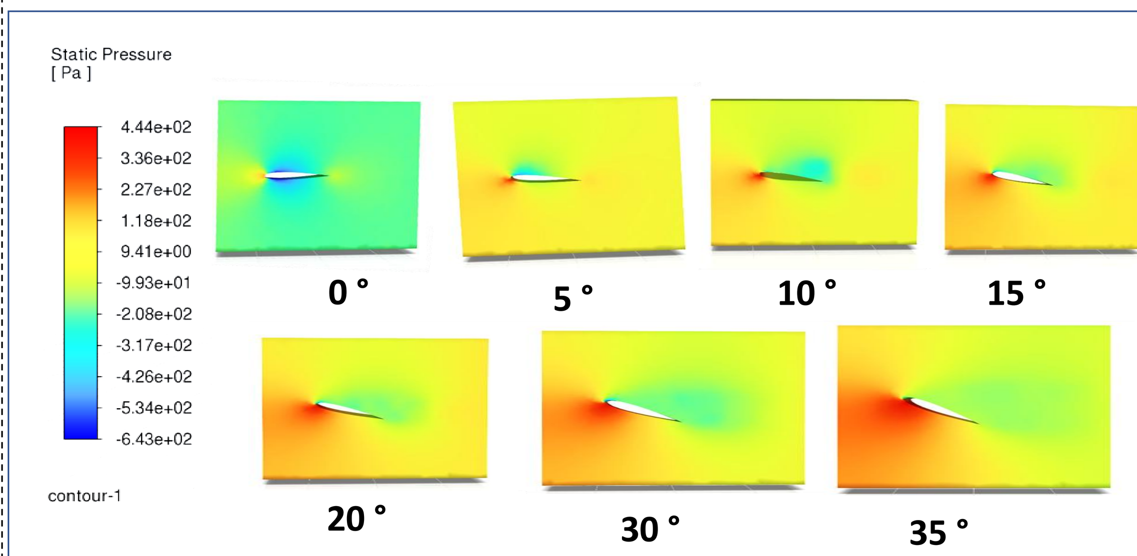


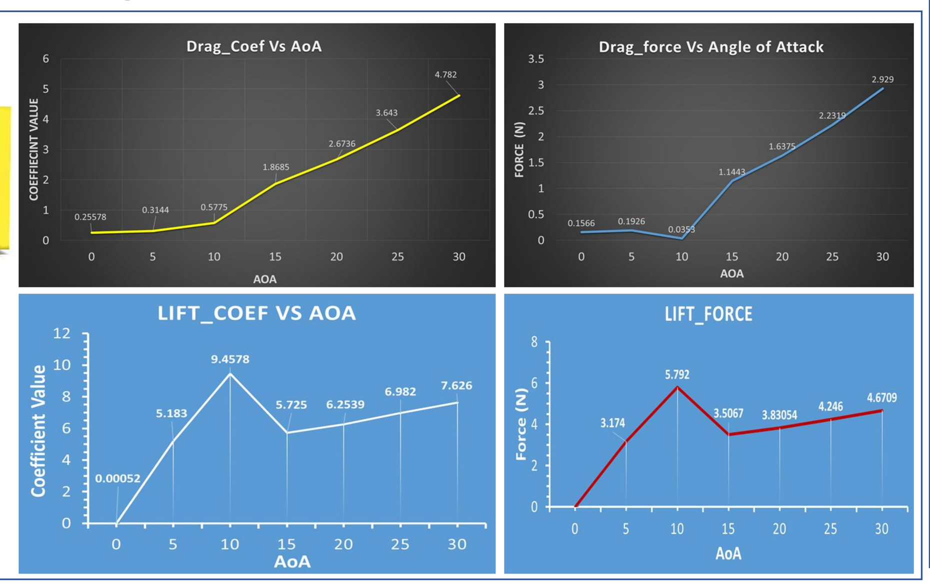
Dynamic Thrust Measurement of Foil



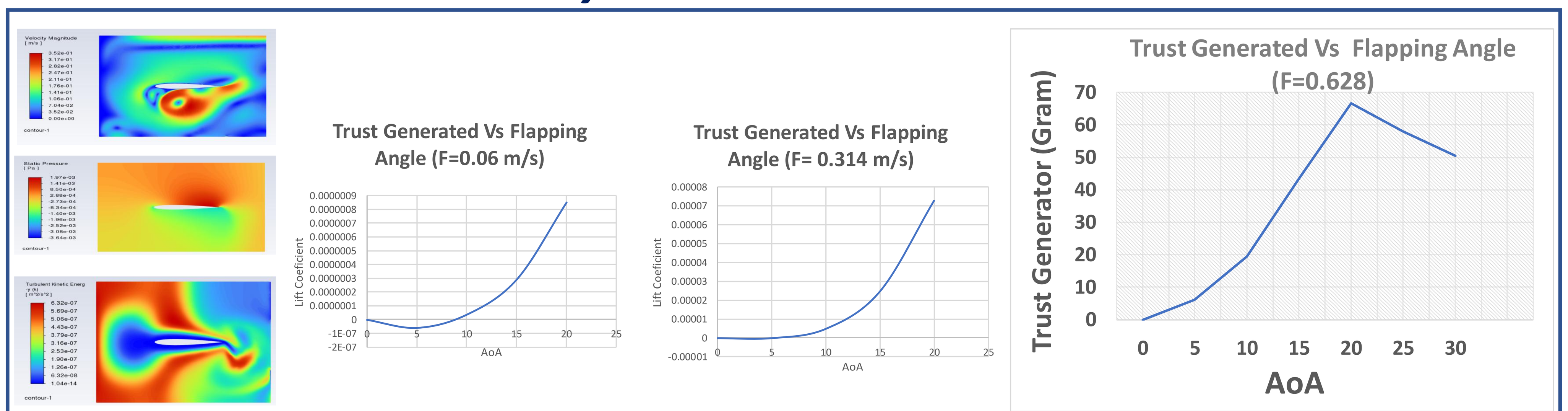
Experimental / Simulation Results

Static Simulation Analysis

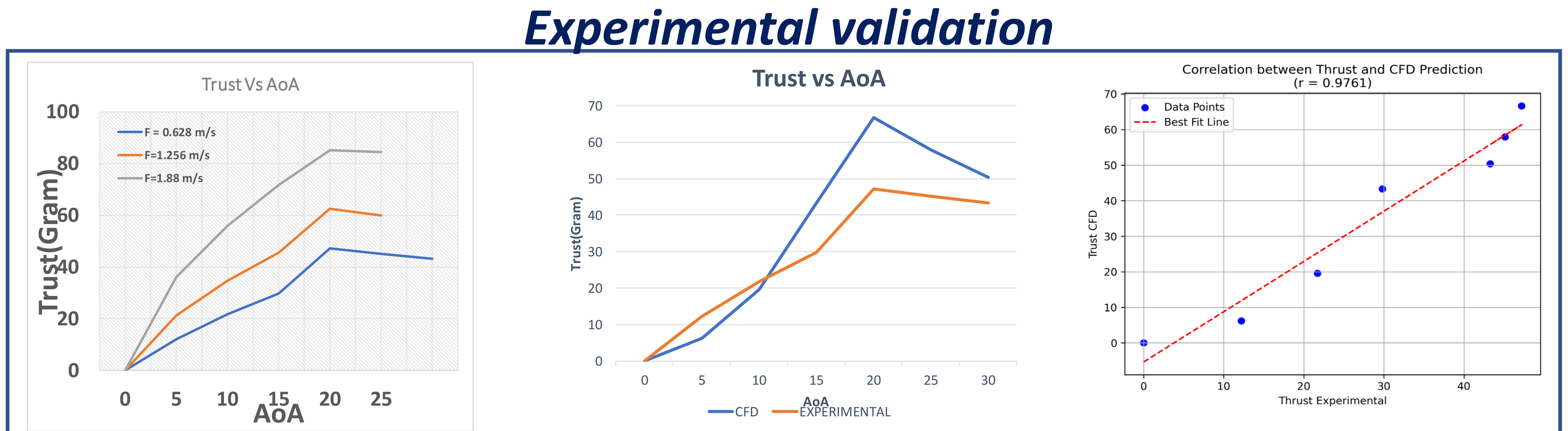




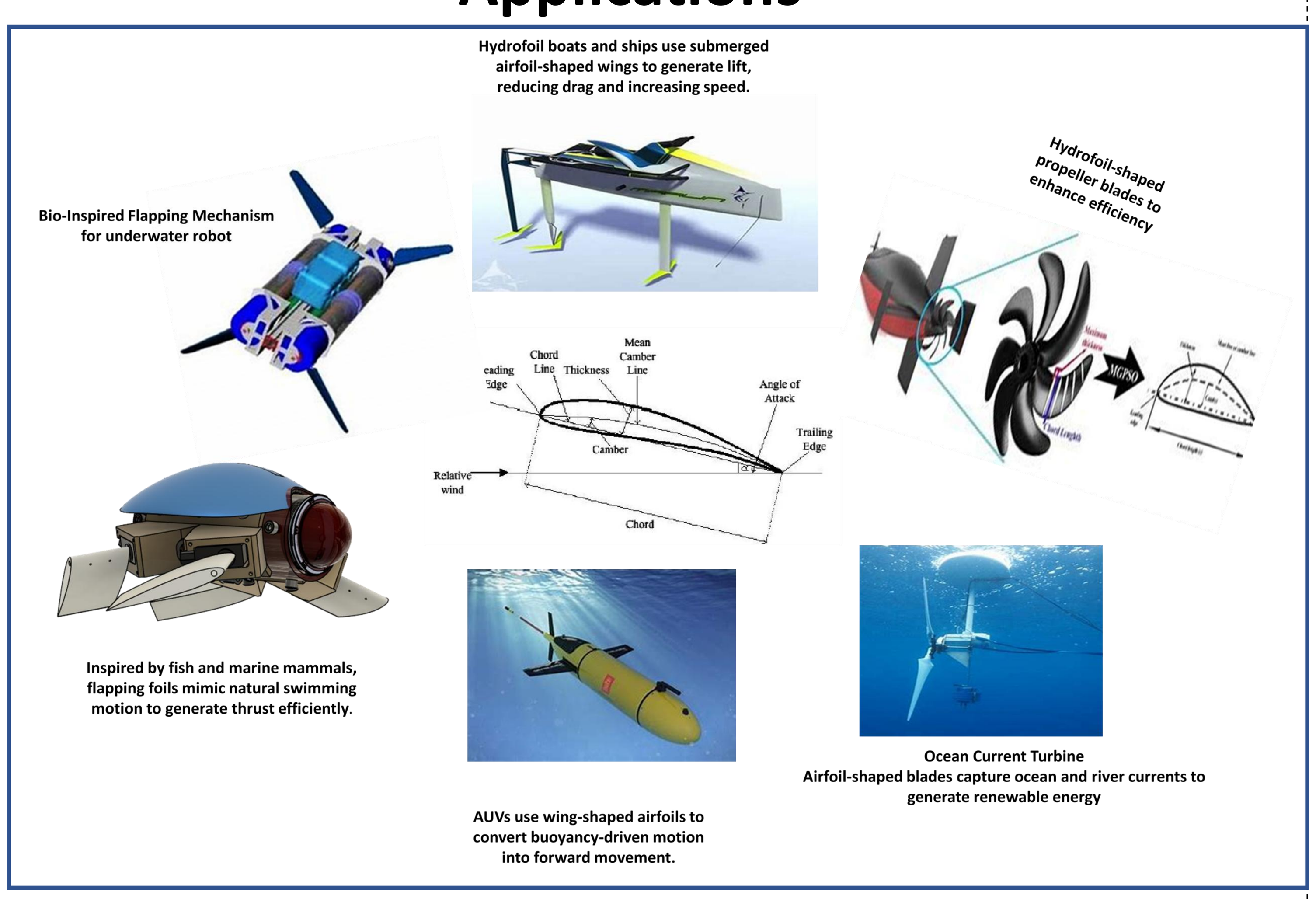
Dynamic Simulation



Experimental validation



Applications



Conclusion

- The correlation analysis between the experimentally measured thrust and the CFD-simulated thrust values for the flapping foil experiment reveals a strong positive correlation (r ≈ 0.976).
- This indicates that the numerical simulations accurately capture the thrust generation characteristics of the flapping foil.
- The close agreement between experimental and computational results validates the CFD model's accuracy, confirming its reliability for predicting hydrodynamic performance.

Future scope

Future studies can further refine the model by incorporating additional flow complexities such as turbulence effects, flexible foil dynamics, and three-dimensional wake interactions to enhance predictive accuracy.

Acknowledgement

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